

Tornado Severity Classification with a Neural Network in TensorFlow

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September 3, 2026

1. Problem and Dataset

This project asks whether a tornado's EF rating can be predicted from when and where it happened and the size of its path. An EF rating is a damage estimate assigned by a surveyor after the event, so the model tests how much of a rating comes from the tornado's geometry and how much from what it hit.

The US Tornado Dataset 1950-2025 contains 73,458 rows and 17 columns with no missing values (Brun, n.d.). I dropped 1,546 unrated rows, leaving 71,912 rated tornadoes, and combined EF2 through EF5 into EF2+ because EF3, EF4, and EF5 each fall below the 5% class-size rule. Always predicting EF0, the largest class in Figure 1, gives a majority-class accuracy of 0.4634 on both validation and test data. Figure 1 also shows the rating mix changing sharply by decade, which is why year was not used as an input.

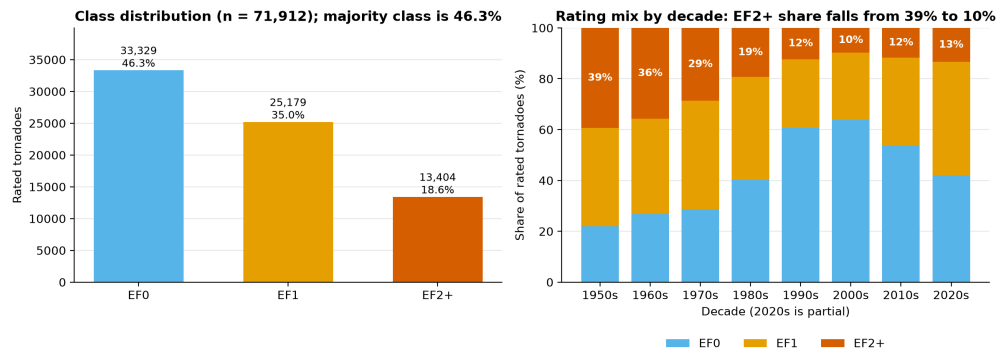


Figure 1. Class distribution and rating mix by decade

The network receives 61 inputs: start latitude and longitude, log-transformed path length and width, sine and cosine pairs for month and hour, and 53 one-hot state indicators. The output is a three-value probability vector for EF0, EF1, and EF2+. Injuries, fatalities, loss, and end coordinates were excluded, the first three because they leak the label and the last because a third are recorded as zero. Figure 2 maps the start points over state boundaries (U.S. Census Bureau, 2024); the EF2+ share is highest in the Plains and Southeast.

Path length and width are strongly right-skewed, so I applied $\log(1 + x)$, and the continuous features were standardised with a scaler fit only on the training data. The data was split with stratification into 50,338 training, 10,787 validation, and 10,787 test rows.

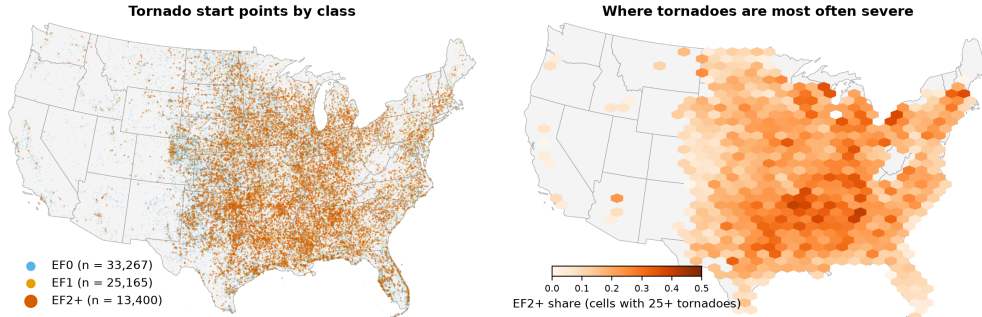


Figure 2. Tornado start points by class and EF2+ share across the contiguous US

2. Model Design

The baseline is a fully connected network with 61 inputs, two hidden Dense layers of 64 ReLU units each, and a three-unit softmax output, for 8,323 trainable parameters (Table 1). Two hidden layers can learn interactions such as a long, wide track in the Plains, and 64 units per layer is a modest starting point before varying capacity in Section 3.

Baseline: 61 inputs, two hidden layers of 64 ReLU units, 3-way softmax output; Adam (lr 0.001), sparse categorical cross-entropy, 30 epochs, batch size 32.

layer	type	units	activation	params
dense	Dense	64	relu	3,968
dense_1	Dense	64	relu	4,160
dense_2	Dense	3	softmax	195

Table 1. Baseline architecture

I compiled the model with Adam at its default learning rate of 0.001, sparse categorical cross-entropy for integer labels, and accuracy as the metric. Each model trained for 30 epochs with a batch size of 32 and validation data supplied to `fit`. TensorFlow, NumPy, and Python seeds were set at the top of the notebook and reset before each run.

3. Experiments

I ran the baseline and seven controlled experiments with the same data, seed, and 30-epoch training period, each changing one factor. For activation, I replaced ReLU with tanh and sigmoid to test whether saturating activations mattered. For the optimizer, I replaced Adam with SGD and RMSprop to test whether adaptive step sizes mattered. For capacity, I compared narrow 16-16 layers with wide 256-256 layers. The wide model overfit the most, so I added dropout of 0.3 to that

architecture.

Table 2 ranks all runs by final-epoch validation accuracy. With 10,787 validation rows, the standard error of an accuracy near 0.61 is about 0.005, so differences under one percentage point are noise.

All runs ranked by final-epoch validation accuracy (best in bold). Binomial s.e. of a validation accuracy is about 0.005, so differences under one point are within noise.

run	category	changed	train_acc	val_acc	gap
wide+dropout	regularisation	wide + dropout 0.3	0.6341	0.6161	0.0180
opt=rmsprop	optimizer	adam -> rmsprop	0.6333	0.6125	0.0208
act=sigmoid	activation	relu -> sigmoid	0.6160	0.6118	0.0043
opt=sgd	optimizer	adam -> sgd	0.6165	0.6113	0.0052
narrow	capacity	64,64 -> 16,16	0.6199	0.6095	0.0104
act=tanh	activation	relu -> tanh	0.6306	0.6094	0.0212
baseline	baseline	-	0.6448	0.6079	0.0369
wide	capacity	64,64 -> 256,256	0.7056	0.5939	0.1117

Table 2. All runs by validation accuracy

4. Results and Evaluation

Figure 3 shows the baseline learning curves: validation accuracy plateaus near 0.61 by epoch 3 while training accuracy keeps rising.

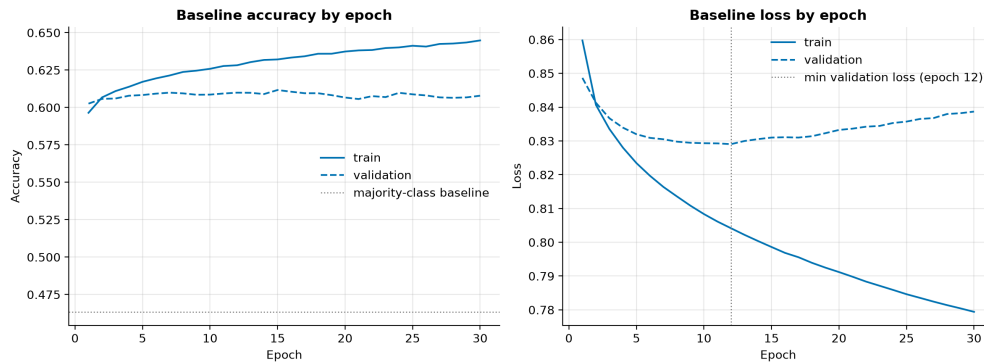


Figure 3. Baseline training and validation curves

Figure 4 compares validation accuracy across the experiment groups. The wide model peaks early and then declines as it memorises the training data, while dropout keeps the same architecture above the baseline for most of training. Wide plus dropout finished first (Table 2).

I evaluated the wide-plus-dropout model once on the 10,787 test rows, which were split off first and untouched until this step. It reached 0.6130 accuracy against the majority baseline of

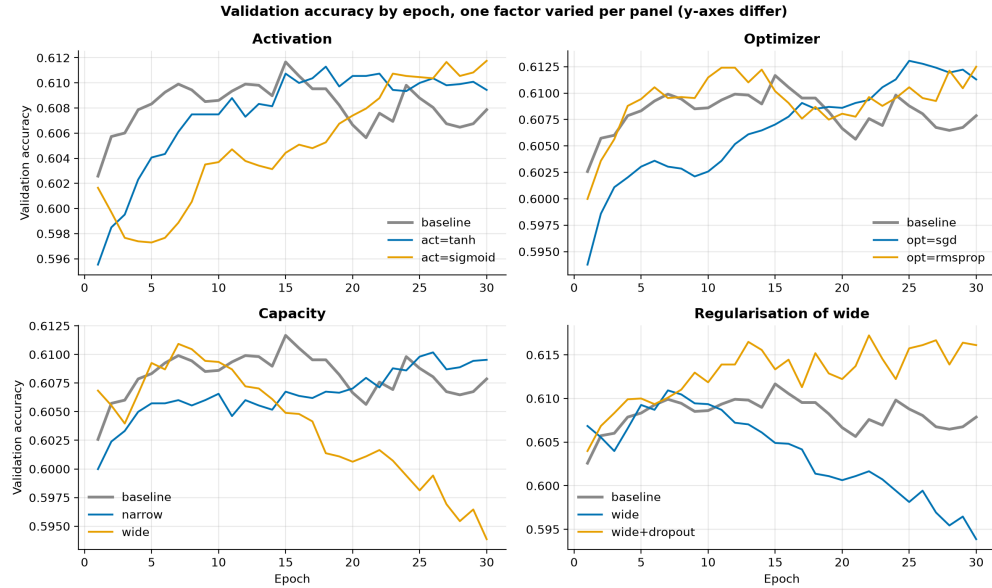


Figure 4. Validation accuracy by epoch for each experiment group

0.4634, an absolute improvement of 0.1495.

Accuracy hides how uneven the model is across classes (Figure 5). Most errors are between adjacent classes: half of true EF2+ tornadoes are predicted as EF1, and almost no EF0 tornadoes are predicted as EF2+. The model is conservative about calling a tornado severe, with reasonable EF2+ precision but recall of only 0.33. One explanation is rural under-rating: a strong tornado over open land leaves a long, wide track but little damage to inspect, so it receives a lower rating (Doswell & Burgess, 1988).

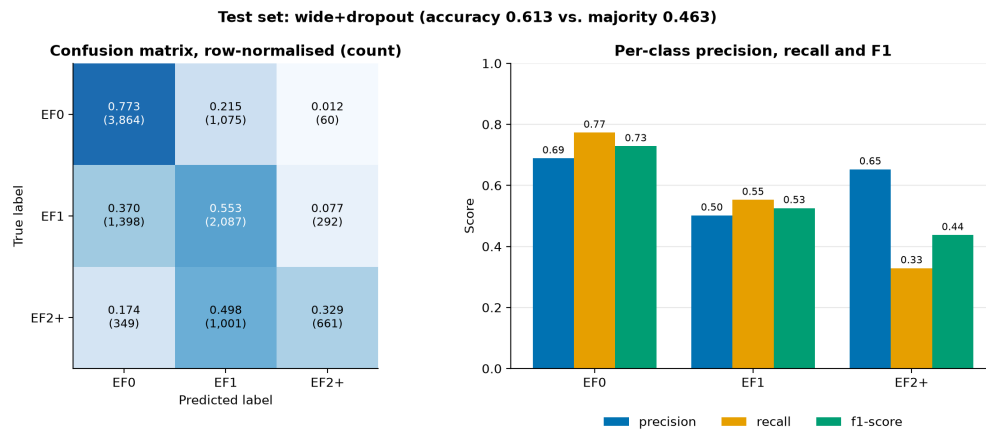


Figure 5. Test-set confusion matrix and per-class metrics

5. Analysis and Interpretation

Wide plus dropout won. The 256-unit layers had the capacity to fit the training data but overfit badly without regularisation; dropout closed most of that gap while keeping some benefit from the extra capacity. However, its margin over the baseline was 0.0083, about 1.8 standard errors, and RMSprop and sigmoid finished within one standard error of it, so “best” means the top of a close group, not a clear victory.

The baseline mildly overfits: in Figure 3, training accuracy continues upward after validation accuracy flattens, and validation loss turns upward after epoch 12. The curves are not both low and still rising, so this is not underfitting and training longer would not help. Since narrow, wide, and regularised models all settle near 0.61, the limit appears to be the inputs rather than the network.

The final model clearly beats the majority baseline on the test set, 0.6130 against 0.4634, or 32% relative improvement. The improvement over the plain two-layer model is less than one percentage point.

The largest problem is that the tornado record changes over time. Figure 1 shows the EF2+ share falling from about 39% in the 1950s to about 10% after 2000, as radar coverage and reporting added many weak tornadoes that would once have been missed (Verbout et al., 2006). A leakage probe shows how much this matters: adding year raises validation accuracy from 0.6079 to 0.6581, more than any legitimate model change. That gain reflects older records skewing strong, not a better geometry-to-rating relationship, which is why year was excluded.

The evaluation has limits as well. A random split mixes every era into training, validation, and test data, so it cannot show whether the model transfers across time, and each configuration was run with one seed. With another week, I would first try a time-based split, training before 2010 and testing after, to see whether the relationship holds for modern records. I would also add land use along each track to test the rural under-rating hypothesis directly.

References

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